

Star Galaxy Quasar Classification using Machine Learning

Ninad Shinkar¹, Anurag Shimpi², Shubham Waghmare³, Varsha Atul Shukre⁴ and Sachin Bhoite⁵

¹⁻⁵Department of Computer Science and Applications, Dr. Vishwanath Karad MIT World Peace University, Pune, Maharashtra

Email: npbshinkar@gmail.com, anuragshimpi91@gmail.com, shubhamwaghmare165@gmail.com, varsha.sontakke@mitwpu.edu.in, sachintenjuly@gmail.com

Abstract—Aim is to create a system that can precisely and efficiently classify stars and galaxies, which can at that point be utilized for different applications in astronomy and astrophysics. The objective of this project is to create a progressed framework for categorizing stars, galaxies, and quasars, while also evaluating various classification models to identify the most effective one. My hypothesis centered on the possibility of preparing a machine learning model to precisely classify stars, quasars and galaxies utilizing the dataset given by the Sloan Digital Sky Survey (SDSS) [25]. We aimed to accomplish ideal execution with balanced data sets. We have utilized machine learning calculation like Support Vector Machine (SVM), Random Forest (RF), Logistic Regression (LR), Decision Tree (DT). The proposed machine learning show can help astronomers in precisely classifying celestial bodies in the sky, possibly contributing to headways in our understanding of the universe. The proposed machine learning show can help astronomers in precisely classifying celestial bodies in the sky, possibly contributing to headways in our understanding of the universe. This research social suggestions have a huge effect on the logical community, not fair in space science but also in other zones that require to classify things accurately.

Index Terms— Data analysis, stars, galaxy, quasar, Machine learning, redshift, Smote.

I. INTRODUCTION

Astronomy is the study of things in space like stars, planets, comets, and systems, as well as stuff like blasts and light appears that happen outside our atmosphere. Stars: These are truly huge, shinning balls of hot gas that remain together since of gravity. They sparkle since of atomic reactions happening inside them. Galaxies: Imagine a huge city of stars, gas, dust, and invisible stuff called dark matter, all held together by gravity. Galaxies can see like spirals, ovals, or indeed fair an enormous messy bunch of stars. Quasars: These are super shinning and capable objects found at the centre of a few universes. They get their energy from huge dark holes sucking in stuff around them.[26]. Our proposal involves the development of a robust classification system capable of accurately categorizing different sources, counting subclasses of galaxies, and recognizing between stars and quasars exclusively based on photometric information. The capacity to separate between stars and quasars and galaxies is crucial for astronomers in distinguishing different astronomical objects [4]. The application of machine learning in star, system, and quasar classification has essentially improved the effectiveness, exactness, and depth of our understanding of these celestial substances, cultivating new discoveries and progressing our information of the universe.

II. LITERATURE REVIEW

The capacity to separate between stars and quasars and galaxies is crucial for astronomers in distinguishing different astronomical objects. [4]. The utilize of machine learning in classification has led to optimized observational methodologies and the extraction of profitable logical results from large-scale sky surveys.[4].

Machine learning models, including Logistic Regression(LR) and Decision Trees(DT), were constructed and trained using data from the Sloan Digital Sky Survey(SDSS)[25].When comparing the execution of the choice tree and multinomial calculated relapse models, the classification report appears that the choice tree show beats the multinomial calculated relapse demonstrate by a critical margin. Achieving 99% accuracy.[26].

Achieved score of 97% for quasars and 99.9% for galaxies. Best classification utilizing blended earlier for high scope sources.[27].

In this article, the authors introduced a Convolutional Neural Network (CNN) model and achieved score of 92.44% for the star system dataset. The CNN model, which automatically extracts features, demonstrated superior classification accuracy compared to the Random Forest (RF) model, achieved score of 78%.[28].

III. ABOUT DATASET

The dataset comprises 10,000 observations of space captured by the SDSS [24]. Each perception is characterized by 17 include columns and 1 target column, showing whether it is a star, galaxy, or quasar. In astronomy, stellar classification involves categorizing stars based on their spectral characteristics. This dataset target is to classify stars, systems, and quasars based on their qualities. The information comprises of 100,000 observations taken by the SDSS, with each observation described by attributes. The 'class' attribute serves as the response variable, which the models aim to predict. Redshift, represented by the 'redshift' property, refers to the phenomenon where light or other electromagnetic radiation from an protest is moved towards the red conclusion of the range, indicating motion away from the observer.[4].

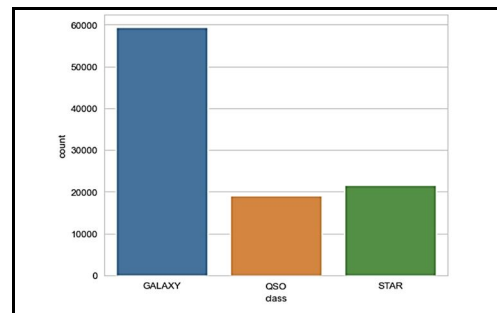


Fig 1: Count Plot of each class

Figure 1 shows the count plot of each class. The motive is to check whether dataset is balanced.[26].

IV. RESEARCH METHODOLOGY

For our project, we use machine learning algorithms including RF, SVM, DT,LR, and XGBoost. After reviewing several research papers, we opted to use these four algorithms randomly. When working on projects involving stars, galaxies, and quasars, employing different machine learning algorithms aids in analysing and comprehending the data. Here is simple overview of some of these algorithms.:

A. Random Forest (RF)

Random Forest is a machine learning algorithm suitable for both classification and regression. It operates by constructing a multitude of DT, forming an ensemble learning approach [9]. In the context of our project, Random Forest can effectively classify various celestial objects based on their distinct features and characteristics [9].

B. Support Vector Machine (SVM)

SVM is a machine learning algorithm used for classification, regression, and outlier detection. SVM, a supervised learning algorithm, works on the principle of finding a hyperplane that optimally separates different classes of data. In our project, SVM proves useful in categorizing and distinguishing between different celestial objects based on their unique properties.

C. Logistic Regression (LR)

LR is a statistical machine learning algorithm used for classification tasks, where the objective is to predict whether a data sample belongs to one of two classes, such as "spam" or "not spam," "true" or "false," or "positive" or "negative." In the context of our project, LR can be employed to predict and categorize celestial objects based on specific characteristics and features [1].

D. Extreme Gradient Boosting (XGBoost)

The XGBoost algorithm is a popular machine learning method utilized for classification tasks, such as distinguishing between stars, galaxies, and quasars. Each of these algorithms has its strengths and can be valuable in analyzing and understanding the vast amount of data related to stars, galaxies, and quasars [4]

Classifier execution is evaluated using Precision (P), Recall (R), and F1-score, which is a consonant cruel of Precision and Recall. The metrics are calculated as follows:

$$P = TP / (TP + FP), R = TP / (TP + FN)$$

where TP is the number of correctly classified positives, FP is the number of incorrectly classified positives, and FN is the number of incorrectly classified negatives [1]. Additionally, the area under the curve (AUC) of the Receiver Operating Characteristic (ROC) curve is used for assessing classifier performance. In ROC curves, the true positive rate (TPR) or Recall is plotted against the false positive rate (FPR) at different threshold values [19].

V. MODEL SETUP

We have three machine learning model for accuracy. Our dataset is divided into three classes named as Star, Galaxy and Quasar. [26].

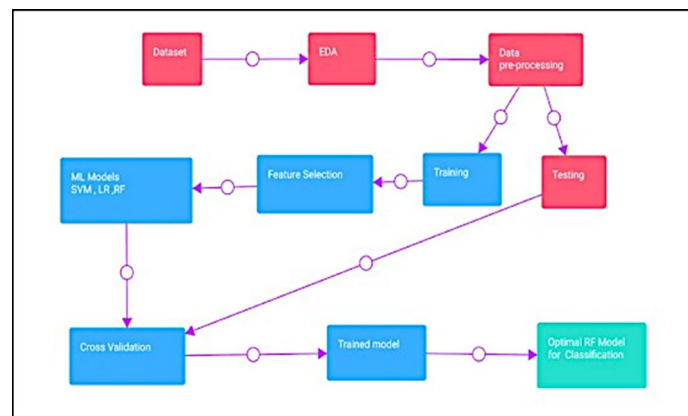


Fig.2: Flowchart of machine learning algorithm

The Fig 2 shows the flowchart that is divided into three main sections: EDA, Training, and Training Model. The steps are labelled with different colours, including purple, blue, and red, to indicate the various stages of the process. The flowchart is designed to show the progression of a project [1].

VI. FEATURE SELECTION

Redshift is a term used in astronomy to describe how the light emitted by objects in space, like stars, galaxies, and quasars, appears to be shifted towards longer wavelengths.[9]. The application of scaling, encoding, SMOTE sampling, and model fitting in machine learning projects for stars, galaxies, and quasars is crucial for preparing and analysing astronomical datasets, addressing class imbalances, and developing accurate models to classify, predict, and understand celestial objects.[9].

Scaling: When we scale data in machine learning, we are basically making sure that all the numbers we use are on a similar scale. This is important because in astronomy, the numbers we get from different measurements can be really big or really small.

Sampling with SMOTE: Sometimes in astronomy, we don't have the same amount of information about all the different types of objects in space. SMOTE is a way to make sure we have a fair amount of information about all the different types, even if some of them are rarer than others.[7].

Model Fitting: When we train a machine learning model, it's like teaching the computer to recognize patterns in the data. In astronomy, we can use different kinds of models, like decision trees or neural networks, to help us make predictions about stars, galaxies, and quasars based on the information we have.[7].

VII. CLASSIFICATION RESULTS

Random forest (RF) is a machine learning algorithm that can be utilized for classification tasks, such as recognizing between stars, galaxies, and quasars in astronomical datasets. During the training process, random forest can handle class imbalances, which is basic when managing with astronomical datasets where a few classes (e.g., quasars) are much rarer than others (e.g., stars). By learning from the majority and minority classes, the calculation can still give precise predictions even when the number of occasions of certain classes is limited.[26].

```
In [12]: rfc = rfc_clf.fit(X_train_smote, Y_train_smote)
print(rfc)

RandomForestClassifier(random_state=42)

In [13]: Y_predicted = rfc.predict(X_test)
score = rfc.score(X_test, Y_test)
rfc_score = np.mean(score)
print('Accuracy : %.3f' % (rfc_score))
print('Accuracy percentage : ' + " {:.2f}%".format(np.mean(score) * 100))

Accuracy : 0.974
Accuracy percentage : 97.42%

In [14]: new_data = np.array([135.689107, 32.484632, 23.87882, 22.27538, 20.39581, 19.16573, 18.79371, 0.634794])
Y_predicted = rfc.predict(new_data)
print(Y_predicted)

['GALAXY']
```

Fig 3: Random Forest algorithm accuracy prediction

```
In [27]: #Model and Accuracy Testing:-
predicted = svm.predict(X_test)
score = svm.score(X_test, Y_test)
svm_score = np.mean(score)
print('Accuracy : %.3f' % (svm_score))
print('Accuracy percentage : ' + " {:.2f}%".format(np.mean(score) * 100))

Accuracy : 0.957
Accuracy percentage : 95.68%
```

Fig 4: SVM algorithm accuracy prediction

The fig 3 represents the model training of Random Forest algorithm and received accuracy of 97.42 which is highest. We have taken a sample of data to predict the class which is the galaxy. [19]. Support Vector Machine (SVM) is a sort of supervised machine learning algorithm that can be utilized for classification tasks, including classifying stars, galaxies, and quasars. SVMs work by finding a hyperplane that maximally isolates the data focuses of distinctive classes whereas minimizing the number of misclassified points. SVMs can offer assistance distinguish the fundamental physics and characteristics of stars, galaxies, and quasars by learning the connections between their features and class labels.[26] The Fig 4 explains the model training and accuracy of Support Vector Machine (SVM) algorithm, the accuracy is 95.68 % [26] Logistic regression (LR) is a popular machine learning algorithm utilized for classification assignments, including classifying stars, galaxies, and quasars. It works by learning the connections between highlights and course names, which helps in predicting the probability of a given perception belonging to a specific class. logistic regression can offer assistance identify important features that recognize these objects from one another. One of the key preferences of logistic regression is its capacity to handle class imbalances, which is a common issue in astronomical datasets.[26].

```
Y_predicted1 = LR_model.predict(X_test)
score = LR_model.score(X_test, Y_test)
LR_score = np.mean(score)
print('Accuracy : %.3f' % (LR_score))
print('Accuracy percentage : ' + " {:.2f}%".format(np.mean(score) * 100))

Accuracy : 0.943
Accuracy percentage : 94.34%
```

Fig 5: Logistic Regression algorithm accuracy prediction

```
: accuracy = accuracy_score(y_test, y_pred)
print("Accuracy: {:.2f}%".format(accuracy * 100))

Accuracy: 93.20%
```

Fig 6: XGboost algorithm accuracy prediction

The figure 5 represents the model training of the Logistic regression algorithm, we have used smote features for this algorithm and the accuracy of this algorithm is 94.34.[19]. XGBoost is a prevalent machine learning algorithm known for its effectiveness in taking care of huge datasets and accomplishing high precision. The Figure 6 represents the model training and accuracy of the XGboost algorithm, it has achieved accuracy of 93.20%.[1].The confusion matrix is a fundamental tool in assessing the execution of a classification algorithm. It gives a tabular representation that summarizes the model's expectations against the real course labels in the

dataset, precision, recall, and other performance metrics. In this case, the algorithm is classifying stars, galaxies, and quasars. The framework is isolated into four quadrants, each representing the number of accurately and incorrectly classified objects. (19).

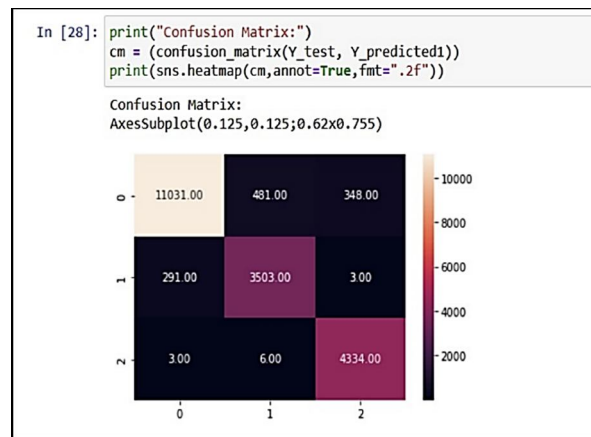


Fig 7: Heat map of confusion matrix

The Figure 7 illustrates the four quadrants commonly found in a confusion matrix used for evaluating classifier performance [4]. Each quadrant represents a specific classification outcome:

True Positive (TP): This quadrant shows the number of stars that were accurately recognized as stars.

False Positive (FP): The top-right quadrant indicates the number of galaxies that were mistakenly classified as stars.

False Negative (FN): The bottom-left quadrant represents the number of quasars that were erroneously identified as stars.

True Negative (TN): Lastly, the bottom-right quadrant displays the number of quasars that were correctly identified as quasars.

Understanding these quadrants helps in assessing the performance and accuracy of the classification model.[26]

VIII. CONCLUSION

In conclusion, star-galaxy-quasar classification is a critical issue in astronomy and astrophysics, with applications in galaxy advancement, cosmology, and extragalactic astronomy. Machine learning algorithms, including Random Forest (RF) and Support Vector Machine (SVM), have been effectively utilized to address the classification of stars, galaxies, and quasars, yielding high levels of accuracy and performance. The application of machine learning techniques to classify celestial objects has been a significant area of interest in astronomical research. The usage of Random Forest algorithm has accomplished an accuracy of 97.42. Comparison of machine learning models, such as Logistic regression (LR), XGBoost and Support Vector Machine (SVM) has too great results.

	precision	recall	f1-score	support
0	0.98	0.95	0.96	11860
1	0.93	0.93	0.93	3797
2	0.93	1.00	0.96	4343
accuracy			0.96	20000
macro avg	0.95	0.96	0.95	20000
weighted avg	0.96	0.96	0.96	20000
SVM model accuracy_score on test dataset: 0.9568				

Classification Report:				
	precision	recall	f1-score	support
GALAXY	0.97	0.93	0.95	11860
QSO	0.88	0.92	0.90	3797
STAR	0.93	1.00	0.96	4343
accuracy			0.94	20000
macro avg	0.93	0.95	0.94	20000
weighted avg	0.95	0.94	0.94	20000

Fig 8: Classification Report of SVM algorithm

Fig 9: Classification Report of Logistic Regression (LR) algorithm

The fig 8 shows the classification report of Support Vector Machine algorithm predicting their accuracy using precision, recall, f1-score. The fig 9 shows the classification report of Logistic Regression (LR) algorithm predicting their accuracy using precision, recall, f1-score.

	precision	recall	f1-score	support
GALAXY	0.97	0.98	0.98	11889
QSO	0.96	0.92	0.94	3792
STAR	0.99	1.00	0.99	4319
accuracy			0.97	20000
macro avg	0.97	0.97	0.97	20000
weighted avg	0.97	0.97	0.97	20000

Fig 10: Classification report of Random Forest algorithm

	precision	recall	f1-score	support
Galaxy	0.95	0.91	0.93	1006
QSO	0.91	0.95	0.93	994
Star	0.00	0.00	0.00	0
micro avg	0.93	0.93	0.93	2000
macro avg	0.62	0.62	0.62	2000
weighted avg	0.93	0.93	0.93	2000

Fig 11: Classification report of XGBoost algorithm

The fig 10 shows the classification report of Random Forest algorithm predicting their accuracy using precision, recall, f1-score.[29] The figure 11 shows the classification report of XGboost algorithm predicting their accuracy using precision, recall, f1-score.[1].

The figures above present a comparison between the four models. The classification report serves as a valuable tool for machine learning algorithms, offering a comprehensive evaluation of a classification model's performance.

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